Flutter Mock Test

Instructions

- Time: **3 Hours**
- Attempt all questions
- Use **Flutter + Dart**
- Use **setState / basic state management only**

Problem Statement

Build an **Employee Management System using Flutter and Firebase Firestore**.

The app should allow users to **add, view, update, and delete employee records** with real-time updates.

Screens Required

1. Home Screen (Employee List)

UI Components:

- AppBar → Title: **Employee Management**
- Text → **Total Employees Count**
- ListView → Display all employees
- FloatingActionButton → **Add Employee**

Employee Data Model

Each employee should contain:

- Name (String)
- Position (String)
- Salary (Number)

Store data in **Cloud Firestore**

Example collection:

```
employees (collection)
  └── employeeId (document)
        ├── name
        ├── position
        ├── salary
```

+ Add Employee

UI:

- TextField → Name
- TextField → Position
- TextField → Salary
- Button → Save

Functional Requirements:

- Validate:
  - All fields must not be empty
  - Salary must be a number
- After adding:
  - Navigate back to Home Screen
  - Data should update automatically

Display Employees

UI:

- Use **ListView.builder**

Each item should show:

- Name
- Position
- Salary

Functional Requirements:

- Fetch data using: streambuilder

Update Employee

UI:

- Tap on employee → Open Edit Screen
- Pre-fill existing data

Functional Requirements:

After update:

- Navigate back
- Changes reflect instantly

Delete Employee

UI:

- Add delete icon/button in each list item

Functional Requirements:

- Show **confirmation dialog**:

  "Are you sure you want to delete?"

App Behavior

- Adding, updating, deleting should reflect **instantly**
- No need to refresh manually

Loading Handling

- Show loading indicator while fetching data (optional but good)

UI Expectations

- Clean layout
- Proper spacing
- Use Material widgets
- Card/ListTile for each employee

Packages Required

- `firebase_core`
- `cloud_firestore`